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SHORT COMMUNICATION

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hOGG1 gene polymorphism and breast cancer risk: A systematic review and meta-analysis study

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Abstract

To address the effect of *hGGO1* (rs1052133) gene polymorphism on the risk of breast cancer, a meta-analysis was performed. We pooled adjusted odds ratios (OR) as overall and three subgroups (menopausal status, ethnicity, and study setting). In overall analysis, we found a significant association when the model of inheritance was homozygote (pooled OR 1.14; 95% CI 1.01, 1.29). Subgroup analysis showed significant association for homozygote genetic models among postmenopause women (OR 1.23; 95% CI 1.01, 1.49) and Asian population (OR 1.17; 95% CI 1.01, 1.35). This study suggested that the carrier of Ser326Cys polymorphism of *hOGG1*, Cys/Cys vs Ser/Ser, are at higher risk for breast cancer, independent of other hormonal and environmental risk factors.

KEYWORDS

breast cancer, *hOGG1*, meta-analysis, polymorphism, rs1052133

1 | INTRODUCTION

Among many polymorphic genes involved in the pathogenesis of breast cancer, the effect of *hGGO1* (rs1052133) gene involved in DNA repair capacity has been studied many times in different populations with controversial results. While many studies reported a significant association between breast cancer and *hOGG1* Ser326Cys polymorphism,[1,2] there are many studies that failed to detect any association[3,4] or even showed an inverse association.[5]

In order to address the discrepancy in association between the *hOGG1* polymorphism and breast cancer, we conducted a systematic review and meta-analysis of all observational studies to evaluate the association of *hOGG1* Ser326Cys polymorphism and risk of breast cancer.

2 | METHODS

2.1 | Search strategy

We systematically searched PubMed and Web of Knowledge databases from their commencements until January 2016. Our search strategy consisted of terms related to polymorphism (Mesh term "Polymorphism, Single Nucleotide") combined with specific terms for breast cancer (Mesh term "Breast Neoplasms"). In addition, references of retrieved literatures were reviewed to include all relevant citations.

2.2 | Inclusion and exclusion criteria

All case-control studies investigating the association between confirmed breast cancer and *hOGG1* Ser326Cys polymorphism were included in this meta-analysis. Studies with no control group, all type of letters to editors, comments, and editorial, animal studies, case reports, and case series, and studies with occurrence of secondary/metastatic breast cancer or all-cause mortality as the main outcome were excluded.

2.3 | Data collection

Two expert researchers (ZB and ASM), independently performed the review process and extracted the required data using a standardized form. The extracted data from each study included; name of first author, publication date, study design, source of controls (population based or hospital based) and major confounders, genotyping methods, minor allele frequency, menopausal status ethnicity, mean age of cases and controls, and odds ratios (OR) and their reported 95% confidence interval for homozygote, heterozygote, recessive model, and dominant model.

2.4 | Quality assessment

Five methodological components, which might bias the association between *hOGG1* gene polymorphism and risk of breast cancer, were assessed: (1) source of control group, (2) ethnicity, (3) Hardy-Weinberg Equilibrium among controls, (4) menopausal status, and (5) sample size.

2.5 | Statistical analysis

After calculating allelic frequency among controls for each study, observed frequencies of the *hOGG1* Ser326Cys genotype was evaluated for Hardy-Weinberg equilibrium using the X^2 statistic. We divided studies into two categories; (1) studies with adjusted OR for major risk factors, and (2) studies that reported crude ORs. We pooled just the ORs that were adjusted for major risk factors. The pooled ORs were estimated using both fixed and the DerSimonian and Laird random-effect models. Heterogeneity across studies was assessed using Q and I^2 statistics.[6,7] An I^2 value above 75% at a significance level of <0.1 was considered as presence of statistically significant heterogeneity.[8] Publication bias and small study effect were assessed with the Egger's test and contour-enhanced funnel plot.[9] The contour-enhanced funnel plot makes it easier to assess the statistical significance of the hypothetical missing studies. We assessed the measure of association for all genetic inheritance models including (Cys/Cys vs Ser/Ser (homozygote), Cys/Ser vs Ser/Ser (heterozygote), Cys/Cys vs Cys/Ser+Ser/Ser (recessive), and Cys/Cys+Cys/Ser vs Ser/Ser (dominant)). In addition to overall analysis (where all groups were combined), subgroup analysis was performed for menopausal status (post- vs premenopause), ethnicity (Asian or European and North American), and study setting (population vs hospital based studies). All analyses were performed using Stata version 13 (Stata Corp LP, College Station, TX, USA).

3 | RESULTS

3.1 | Selected studies

Primary search of databases retrieved 2424 citations. After final screening of full-texts, 19 publications were fulfilled inclusion and exclusion criteria. Among them, nine with adjusted OR

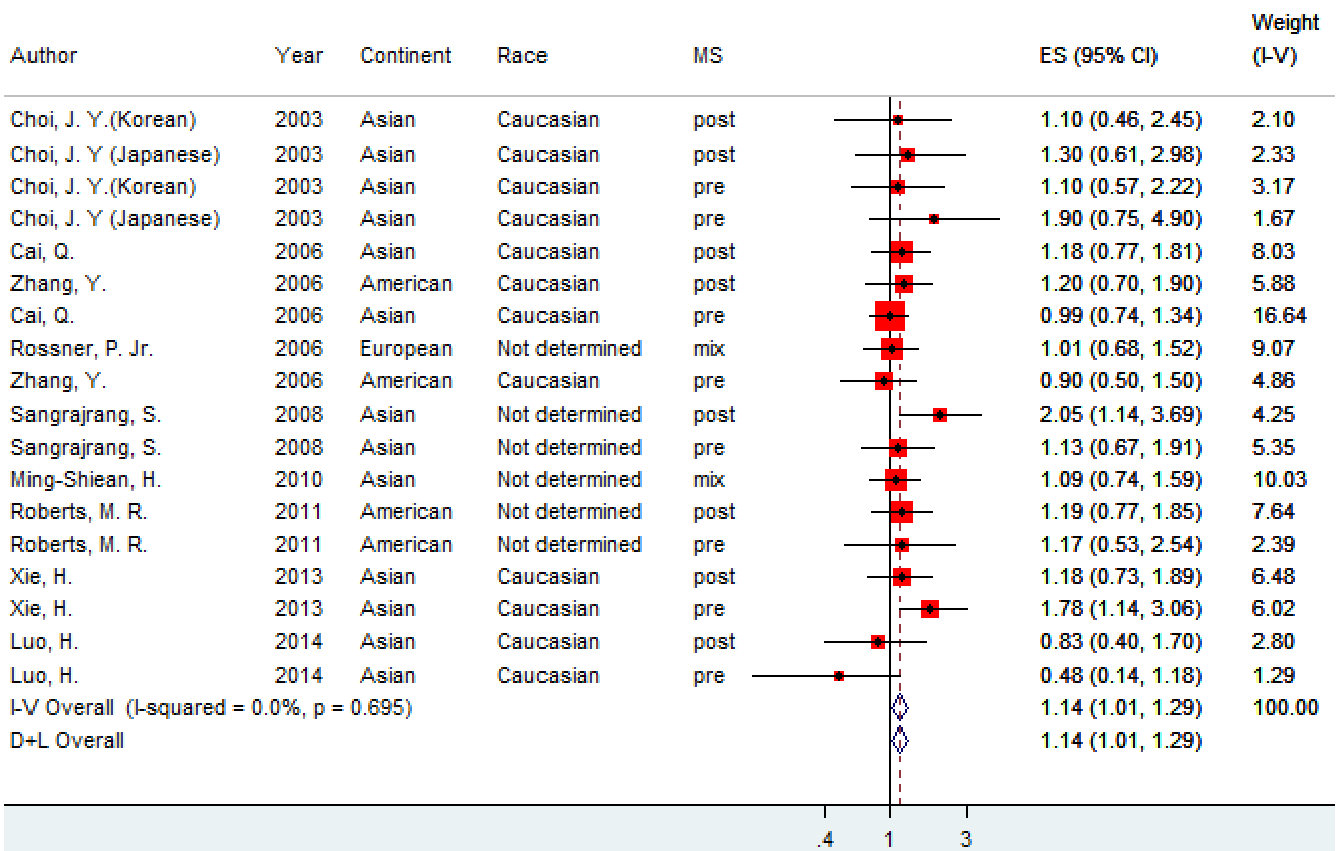

FIGURE 1 Fixed and random-effect model meta-analysis of *hOGG1* Ser326Cys polymorphism for Cys/Cys vs Ser/Ser (homozygote) genetic models and risk of breast cancer. The effect size is odds ratio. MS=menopausal status. [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Subgroup analysis of meta-analysis of *hOGG1* Ser326Cys polymorphism and breast cancer risk

Subgroup	Cys/Cys vs Ser/Ser[a] Studies no.	Fixed model OR (95% CI)	Random model OR (95% CI)	Ph	Ser/Cys vs Ser/Ser[a] Studies no.	Fixed model OR (95% CI)	Random model OR (95% CI)	Ph	Cys/Cys + Ser/Cys vs. Ser/Ser Studies no.	Fixed model OR (95% CI)	Random model OR (95% CI)	Ph	Ser/Cys + Ser/Ser vs. Cys/Cys Studies no.	Fixed model OR (95% CI)	Random model OR (95% CI)	Ph
Overall	18	1.14 (1.01-1.29)	1.14 (1.01-1.29)	0.69	18	1.03 (0.95-1.12)	1.03 (0.95-1.12)	0.99	7	0.97 (0.87-1.08)	0.97 (0.87-1.08)	0.86	4	1.09 (0.91-1.31)	1.01 (0.64-1.58)	0.00
Menopause status																
Post	8	1.23 (1.01-1.49)	1.23 (1.01-1.49)	0.75	8	1.01 (0.90-1.14)	1.01 (0.90-1.14)	0.84	3	0.97 (0.84-1.12)	0.97 (0.84-1.12)	0.95				
Pre	8	1.11 (0.92-1.34)	1.12 (0.90-1.39)	0.37	8	1.04 (0.91-1.20)	1.04 (0.91-1.20)	0.94	2	0.94 (0.70-1.27)	0.81 (0.42-1.56)	0.15				
Design																
Hospital based	13	1.23 (1.05-1.45)	1.23 (1.05-1.45)	0.55	13	0.98 (0.87-1.11)	0.98 (0.87-1.11)	0.99	6	0.97 (0.86-1.11)	0.97 (0.86-1.11)		4	1.09 (0.91-1.31)	1.01 (0.64-1.58)	0.00
Population based	5	1.04 (0.87-1.25)	1.04 (0.87-1.25)	0.90	5	1.07 (0.96-1.19)	1.07 (0.96-1.19)	0.76	1							
Continent																
Asian	13	1.17 (1.01-1.35)	1.17 (1.01-1.36)	0.41	13	1.04 (0.91-1.20)	1.04 (0.91-1.20)	0.95	3	0.96 (0.76-1.16)	0.96 (0.76-1.16)	0.51	2	0.88 (0.70-1.12)	0.88 (0.70-1.12)	0.85
European and American	5	1.08 (0.87-1.35)	1.08 (0.87-1.35)	0.92	5	1.02 (0.92-1.13)	1.02 (0.92-1.13)	0.95	4	0.98 (0.86-1.12)	0.98 (0.86-1.12)	0.94	2	1.57 (1.15-2.13)	1.05 (0.32-3.50)	0.01

OR: odds ratio; CI: confidence intervals; P_h: *P* values of Q-test for heterogeneity test; HB: Hospital-based; PB: Population-based; NM: Not mentioned.
[a]Reference group.

were included into meta-analysis.[3,10–17] One study[17] reported the association for two populations and was considered as two studies.

3.2 | Cys/Cys vs Ser/Ser (homozygote model)

In the overall analysis and in the presence of no significant heterogeneity (I^2: 0.0%, *P*=.69) and no significant small study effect (*b*=0.25, *P*=.71), in the homozygote contrast (genetic model Cys/Cys vs Ser/Ser), a significant association was observed between risk of breast cancer and carrier of the risk allele (326Cys) (OR 1.14 [1.01-1.29]) (Figure 1). Subgroup analysis based on menopausal status showed increase risk only in postmenopausal women (OR 1.23 [1.01-1.49]) but not in premenopausal (OR 1.11 [0.92-1.34]). In study setting, a significant association (OR 1.23 [1.05-1.45]) was observed just in hospital based studies. In ethnicity subgroup analysis, there was a significant association in the homozygote model among Asian subgroup (OR 1.17 [1.01-1.35]).

3.3 | Ser/Cys vs Ser/Ser (heterozygote model)

In this model inheritance, the between-study heterogeneity in the overall and subgroup analyses was not significant (I^2: 0.0%, *P*=.99) and with no small study effect (*b*=−0.16, *P*=.31). Significant increased breast cancer risk was not seen in overall analysis (OR 1.03 [0.95-1.12]). The association was not significant in subgroup analysis based on menopausal status and ethnicity (Table 1).

3.4 | Cys/Cys+Ser/Cys vs Ser/Ser (dominant model)

Considering the dominant model of inheritance, there was no association between the polymorphism and breast cancer (OR 0.97 [0.87-1.08]). There was no heterogeneity (I^2: 0.0%, *P*=.86) and small study effect (*b*=−1.00, *P*=.09) in this analysis. The association remained nonsignificant based in all subgroups (ethnicity, study setting, and menopausal status) (Table 1).

3.5 | Ser/Cys + Ser/Ser vs Cys/Cys (recessive model)

For the recessive model, only four studies reported adjusted OR. There was observable heterogeneity (I^2: 80.2%, *P*=.002) but no evidence for small study effect (*b*=−2.72, *P*=.56) in analysis of this model. In overall category, no association was seen in recessive model (OR 1.01 [0.64-1.58]) (Table 1).

4 | DISCUSSION

Our study demonstrated that there is statistically significant association between the *hOGG1* Ser326Cys polymorphism and breast cancer contrary to many other meta-analyses that failed to report a statistically significant association. Our literature search retrieved three systematic review and meta-analyses that have pooled studies evaluating association between BC and polymorphism Ser326Cys (rs1052133) with none demonstrating a statistically significant association except our study which demonstrated an access risk of 14% among the homozygote carrier of the polymorphism.

The first meta-analysis[18] reported that a statistically significant inverse association favoring carriers for different inheritance models additive model, homozygote contrast and dominant model except recessive model. The next meta-analysis[19] did not detect an inverse association but ruled out any association. The most recent meta-analysis[20] study reported statistically significant association in the subgroup analysis (for Asia population and postmenopausal breast cancer in recessive model of inheritance). Our study updated the previous studies and included 20 studies and showed the association of the polymorphism with breast cancer which maximally adjusted for potential confounders among homozygote carrier.

We exclude studies that report just crude or unadjusted ORs. This could well explain why previous studies were unable to observe a statistically significant association in overall estimates.

5 | CONCLUSIONS

This study suggested that the Ser326Cys polymorphism of *hOGG1*, Cys/Cys vs Ser/Ser, is important risk factor for breast cancer, independent of major risk factors and the fact that this association is significant among Asian and postmenopausal women.

CONFLICT OF INTEREST

No conflict of interest.

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